例子

Contents

1 Topology 拓扑

Example 1.1. *Assume X is an infinite set with cofinite topology. Then it is easy to check that X is noetherian and the only irreducible closed subset of X is itself so that X is of dimension 0. While given any open covering $\mathcal{U}$ of X, we cannot refine it to $\mathcal{V}$ satisfying that there exists n_0 such that any $n_0 + 1$ open subsets in $\mathcal{V}$ have no common element. Conclude that X has covering dimension infinite.*

2 Abstract Algebra 抽象代数

Example 2.1. *Assume $A = F_2$, $B = F_3$, where F_2, F_3 are respectively the free group generated by 2 and 3 elements. Then there exists natural injective homomorphism from A to B*

$$\begin{aligned} f : A &\longrightarrow B \\ a &\longmapsto a' \\ b &\longmapsto b' \end{aligned} \tag{1}$$

also injective homomorphism from B to A

$$\begin{aligned} g : B &\longrightarrow A \\ a' &\longmapsto a^2 \\ b' &\longmapsto b^2 \\ c' &\longmapsto ab \end{aligned} \tag{2}$$

Example 2.2. *$\mathbb{Z}_p$ for a prime number is all abelian simple groups.*

Example 2.3. *For every finite field $\mathbb{F}$, $PSL_n(\mathbb{F}) = SL_n(\mathbb{F})/Z(SL_n(\mathbb{F}))$ is simple group when $n \geq 2$.*

Example 2.4. *A_5 is the smallest nonabelian simple group.*

Example 2.5. *For special linear group, we have*

$$Aut(SL_n(\mathbb{Q})) \cong Aut(PGL_n(\mathbb{Q})) \cong \begin{cases} PGL_n(\mathbb{Q}) \rtimes \{1, \psi\} & n \geq 3 \\ PGL_n(\mathbb{Q}) & n = 2 \end{cases} \tag{3}$$

where ψ is a canonical automorphism of $GL_n(\mathbb{Q})$

$$\begin{aligned} \psi : GL_n(\mathbb{Q}) &\longrightarrow GL_n(\mathbb{Q}) \\ A &\longmapsto (A^{-1})^T \end{aligned} \tag{4}$$

Example 2.6. *If $n \neq 6$, then $Aut(S_n) = Inn(S_n)$. When $n = 6$, there exists an automorphism* *of S_6 which is not inner*

$$\begin{aligned}
\psi : S_6 &\longrightarrow S_6 \\
(1\ 2) &\longmapsto (1\ 2)(3\ 4)(5\ 6) \\
(2\ 3) &\longmapsto (1\ 4)(2\ 5)(3\ 6) \\
(3\ 4) &\longmapsto (1\ 3)(2\ 4)(5\ 6) \\
(4\ 5) &\longmapsto (1\ 2)(3\ 6)(4\ 5) \\
(5\ 6) &\longmapsto (1\ 4)(2\ 3)(5\ 6)
\end{aligned} \tag{5}$$

And we have $Aut(S_6) \cong S_6 \rtimes \{1, \psi\}$

Example 2.7. *Here is an example which is solvable group but not nilpotent group.*

$$G = \begin{pmatrix} \mathbb{C}^\times & \mathbb{C} & \mathbb{C} \\ 0 & \mathbb{C}^\times & \mathbb{C} \\ 0 & 0 & \mathbb{C}^\times \end{pmatrix} \supseteq G^{(1)} = \begin{pmatrix} 1 & \mathbb{C} & \mathbb{C} \\ 0 & 1 & \mathbb{C} \\ 0 & 0 & 1 \end{pmatrix} \supseteq G^{(2)} = \begin{pmatrix} 1 & 0 & \mathbb{C} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \supseteq G^{(3)} = I_3 \tag{6}$$

Example 2.8. *Take $G_0 = D_8$, $G_1 = \langle r^2, s\rangle$, $G_2 = \langle s\rangle$ and $G_3 = \{1\}$. Then $G_0 \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq G_3$ is a composition series which cannot be obtained by refining some normal series.*

3 Algebraic number theory 代数数论

Example 3.1. *L/K are Galois extension of number fields. Then $k(\mathfrak{P})/k(\mathfrak{p})$ is extension of finite fields.*

$$\begin{aligned}
k(\mathfrak{p}) &\cong \mathbb{F}_q, \quad q = N(\mathfrak{p}) = \sharp k(\mathfrak{p}) \\
k(\mathfrak{P}) &\cong \mathbb{F}_{q^f}, \quad f = [k(\mathfrak{P}) : k(\mathfrak{p})]
\end{aligned} \tag{7}$$

Moreover, $Gal(k(\mathfrak{P})/k(\mathfrak{p})) \cong \mathbb{Z}/f\mathbb{Z}$ generated by Frobinius automorphism $\sigma_q : \mathbb{F}_{q^f} \longrightarrow \mathbb{F}_{q^f} \quad x \longmapsto x^q$ which is canonical and natural.

4 Commutative Algebra 交换代数

Example 4.1. *Take $G = \mathbb{Z}[\frac{1}{p}]/\mathbb{Z}$. Then G is Artinian as $\mathbb{Z}$-module but not noetherian.*

Example 4.2. $\mathbb{Z}[\frac{1}{p}]$ *and $k[x_1, x_2, \cdots]$are neither noetherian nor Artinian.*

Example 4.3. *A ring with only one prime ideal may not be noetherian. Let k be a field and $A = k[x_1, x_2, \cdots]$, $I = (x_1, x_2^2, \cdots, x_n^n, \cdots)$. Then the ring A/I has only one prime ideal $(\overline{x_1}, \overline{x_2}, \cdots)$ but not noetherian.*

Remark 4.1. *To show A/I has only one prime ideal, only need to show there is exactly one prime ideal of A containing I.*

Example 4.4. *The only valuations on $\mathbb{Q}$ are the p-adic valuations $|\cdot|_p$ for prime numbers p and the trivial valuation $|\cdot|_0$. Thus there is a one-to-one correspondence between* $\mathrm{Spec}(\mathbb{Z})$ *and $Spv(\mathbb{Q})$.*

Example 4.5. *$Spv(\mathbb{Z}) = Spv(\mathbb{Q}) \cup \{|\cdot|_{0,p} \,\big|\, p$ is prime number$\}$, where $|\cdot|_{0,p}$ is the valuation induced by the trivial valuation on $\mathbb{F}_p$.*

Example 4.6. *Take $\mathbb{Z}$-modules $A' = \mathbb{Z}$, $A = \mathbb{Z}$ and $A'' = \mathbb{Z}/2\mathbb{Z}$. There is an exact sequence $0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z}$, where A' and A are flat but A'' is not.*

5 Algebraic Geometry 代数几何

5.1 Sheaves

Example 5.1 (***Sheaf Hom Doesn't Commute with Taking Stalk***)**.** *Take $X = \mathbb{R}$ as topological space with structure sheaf constant sheaf associated to $\mathbb{Z}$. Consider the constant sheaf associated to $\mathbb{Z}$ over $\mathbb{R} \setminus \{0\}$ and push forward it to X, denoted by $\mathcal{F}$. Then $\mathcal{F}_0 = 0$ while for all open neighbourhood $U \ni 0$, $\mathcal{F}\big|_0 \neq 0$. Thus $\mathrm{Hom}(\mathcal{F}_0, \mathcal{F}_0) = 0$ while $\mathcal{H}om(\mathcal{F}, \mathcal{F})_0 \neq 0$.*

5.2 Affine schemes

Example 5.2. *Take $A = k[x_1, x_2, \cdots]$ and $\mathfrak{m} = (x_1, x_2, \cdots)$. Then $D(\mathfrak{m}) = \cup_i D(x_i)$ is an open subset of $\operatorname{Spec} A$ which is not quasi-compact.*

Example 5.3. *Take R to be a DVR. Assume $\mathfrak{m} = (\varpi)$ is the maximal ideal of R and $k = R/\mathfrak{m}$ is the residue field. Then $\operatorname{Spec} R[x]$ has two maximal ideals of different height, which are $(1 - \varpi x)$ and (x, ϖ).*

Example 5.4. *Take $X = \operatorname{Spec}(k[x, y, z]/(z^2 - xy))$. Then X is normal and noetherian but not regular at $(\overline{x}, \overline{y}, \overline{z})$.*

Example 5.5. *Take $A = \mathbb{Z}[x]/(2x)$. Then $\operatorname{Spec} A$ is connected but not irreducible with two different minimal prime ideals (2) and (x). Moreover, if take $A = \mathbb{Z}[x]/(4x)$, then $\operatorname{Spec} A$ is even not reduced.*

5.3 Spectrum maps

Example 5.6. *The morphism $\operatorname{Spec} \mathbb{Z}_p \longrightarrow \operatorname{Spec} \mathbb{Z}$ is not of finite type.*

Example 5.7. *Let k be a field and $A = k[x_1, x_2, \cdots]/(x_1^2, x_2^2, \cdots)$. Then we have that $A/\mathfrak{N} = k$ which is noetherian. Note that $\operatorname{Spec} A = \operatorname{Spec}(A/\mathfrak{N})$, we get an example that A is not noetherian but spectrum of it is noetherian.*

Example 5.8. *Take $A = k[x, y]/(x^2, xy)$ and $X = Y = \operatorname{Spec} A$, where k is algebraically closed. Take $f = \mathrm{id}_A^*$ and $g = \varphi^*$, where φ is given by $\varphi(x) = 0$ and $\varphi(y) = y$. Then these give a counterexample of Hartshorne exercise ii 4.2(a).*

Example 5.9. *Even though $\mathbb{A}^1_{\mathbb{C}} \to \mathbb{C}$ is separated, of finite type and closed, it is not proper since the natural map $\mathbb{A}^2_{\mathbb{C}} \to \mathbb{A}^1_{\mathbb{C}}$ is not closed mapping $V(xy - 1)$ to set of all closed points except (x). This satisfies our exception because the complex manifold corresponding to $\mathbb{A}^1_{\mathbb{C}}$ is not compact.*

Example 5.10. *Let* $f : \operatorname{Spec}\mathbb{C} \to \operatorname{Spec}\mathbb{R}$ *be the canonical morphism. Consider the base change of* f *by itself, we get* $f \times_{\operatorname{Spec}\mathbb{R}} f : \operatorname{Spec}(\mathbb{C}\otimes_{\mathbb{R}}\mathbb{C}) \longrightarrow \operatorname{Spec}\mathbb{C}$*. Note that* $\mathbb{C}\otimes_{\mathbb{R}}\mathbb{C} \cong \mathbb{C}\times\mathbb{C}$*, we get that* $f \times_{\operatorname{Spec}\mathbb{R}} f$ *is not an isomorphism. Hence bijection and injection are not stable under base change.*

Example 5.11. *Take* $f : \operatorname{Spec} k[t,x,y]/(x^2, tx, xy) \to \operatorname{Spec} k[t]$*. Then all fiber dimension would be* 1*. And the fiber at* $t = 0$ *is* $\operatorname{Spec} k[x,y]/(x^2, xy)$ *not purely dimensional. Hence for a morphism with constant fiber dimension can have not purely dimension fiber at some closed point.*

5.4 Examples with affine line with double origin

Example 5.12. *Take* X *to be affine line and* Y *to be affine line with two origin. Take* f, g *to be two inclusions from* X *to* Y *respectively. Then these give a counterexample of Hartshorne exercise ii 4.2(b).*

Example 5.13. *Take* X *to be affine line and* Y *to be affine line with two origin. Then there is natural morphism* $f : Y \to X$ *and inclusion* $i : X \hookrightarrow Y$*. Note that* f *is not separated and* i *is a section of* f*, which is not closed immersion.*

Example 5.14. *Take* $A = \mathbb{Z}[x]/(3x^2+2x+1)$ *and* $B = \mathbb{Z}$*. Then* $f : \operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(B)$ *is surjective, of finite type and quasi-finite but not integral, hence also not finite.*

5.5 Sheaves of module

Example 5.15. *Consider affine scheme* $X = \operatorname{Spec}(\mathbb{Z})$*. Take* $M = \oplus_{p \text{ prime number}} \mathbb{Z}/p\mathbb{Z} \in Mod_{\mathbb{Z}}$*. Then the corresponding sheaf* $\widetilde{M}$ *is finitely generated locally but not a finitely generated* $\mathcal{O}_X$*-module.*

Example 5.16. $\mathcal{O}_{\mathbb{P}^1} \xrightarrow{\cdot x} \mathcal{O}_{\mathbb{P}^1}(1)$ *is injective as sheaf map but not injective as bundle map.*

5.6 Geometric properties

Example 5.17. *Take* $X = \operatorname{Spec}(\mathbb{R}[x,y]/(x^2+y^2))$ *over* $\operatorname{Spec}\mathbb{R}$*. Then* X *is integral but not geometrically irreducible thus not geometrically integral.*

Example 5.18. *Take* $X = \operatorname{Spec}\mathbb{C}$ *over* $\operatorname{Spec}\mathbb{R}$*. Then* X *is integral but not geometrically connected thus not geometrically integral.*

Example 5.19. *Take* $X = \operatorname{Spec} k(u)$ *over* $\operatorname{Spec} k(u^p)$*, where* k *has characteristic* p*. Then* X *is reduced but not geometrically reduced.*

6 Homological Algebra 同调代数

Example 6.1. *Take* $\alpha = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ *and* $\beta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. *Then* $\alpha\beta = \beta\alpha$. *Take* $G = \langle\alpha\rangle \times \langle\beta\rangle$ *and field* k *of characteristic* p. *Consider* k^3 *as* kG*-module. Then* $\dim_k(Fix^G(k^3)) = 2$ *and* $\dim_k(Cofix_G(k^3)) = 1$. *Thus* $Fix^G \ncong Cofix_G$, *which gives a example that left adjoint and right adjoint (of trivial functor) are not isomorphic.*

7 Representation Theory 表示论

Example 7.1. *Take* $G_1 = D_8$ *and* G_2 *is Quoternions. Then* G_1 *and* G_2 *have the same character tables.*